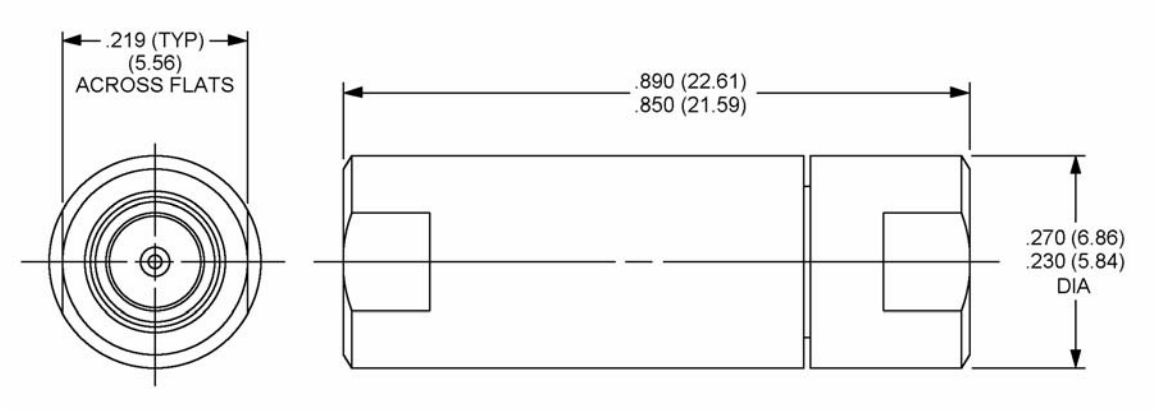
INCH-POUND

MIL-DTL-3933/27
22 December 2004

ATTENUATORS, FIXED, COAXIAL LINE (SMP CONNECTORS - SMP FULL DETENT MALE TO SMP FULL DETENT MALE), FREQUENCY RANGE: DC TO 18 GHZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents, in parentheses, are given for general information only.
3. Unless otherwise specified:
   a. All dimensions are after plating.
   b. Break corners and edges .005 R maximum.
   c. Chamfer first and last threads.
   d. Surface roughness 63 ✓ .
   e. Remove all burrs.
   f. Diameters on common centers to be concentric within .005 T. I. R.
4. Tolerances: Decimals ±.030 (0.76 mm) for one decimal point, ± .015 (0.38 mm) for two decimal points and ±.005 (0.13 mm) for three decimal points.
5. SMP connector interfaces (both ends of attenuator) shall be in accordance with MIL-STD-348, Figure 326-2.

FIGURE 1. Dimensions and configurations for M3933/27-01 through M3933/27-40.

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ENGINEERING DATA:

Attenuators for use in general field equipment.

Materials:

Bodies and sleeves: Stainless steel in accordance with AMS-5640 UNS 530300, type I.

Contact: Beryllium copper in accordance with ASTM-B196/B196M, condition TD04. Alloy UNS-C17300.

Insulators: Teflon in accordance with ASTM-D1710.

Finishes:

Bodies and sleeves: Passivate in accordance with AMS-QQ-P-35C, type II.

Contact: Gold in accordance with ASTM-B488, type II, grade C, class I.27 over nickel plate in accordance with AMS-QQ-N-290, class I.

Operating frequency range: DC to 18 gigahertz (GHz) (see table I).

Operating temperature range: -55°C to +100°C.

Maximum weight: 0.16 ounce.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Nominal impedance: 50 ohms.

Attenuation: see table I. Stability and sensitivity:

Maximum attenuation change after	**Up to 10 dB, inclusive (dB)**	**Over 10 dB (dB/dB)**
Temperature change or thermal shock	0.5	0.05
Vibration or shock	0.5	0.05
Moisture resistance or salt spray	0.5	0.05
Peak power	0.5	0.05

Power: See table I.
Power sensitivity of attenuation, maximum: 0.005 dB/dB/Watt
Temperature sensitivity of attenuation, maximum: 0.0006 dB/dB/°C.

VSWR:

DC to 4 GHz: 1.20:1 maximum.
4 to 12 GHz: 1.35:1 maximum.
12 to 18 GHz: 1.50:1 maximum.

Connector repeatability, maximum variation in attenuation: 0.02 dB.

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TABLE I. Electrical characteristics.

Dash number	Attenuation (dB)		Maximum power input	
	Nominal	Deviation (DC - 18 GHz)	AV (W) at 25°C (continuously)	Peak (*) (W)
01NS	0.5	± .4	0.5	50
02NS	1.0	± .4	0.5	50
03NS	1.5	± .4	0.5	50
04NS	2.0	± .4	0.5	50
05NS	2.5	± .3	0.5	50
06NS	3.0	± .3	0.5	50
07NS	3.5	± .3	0.5	50
08NS	4.0	± .3	0.5	50
09NS	4.5	± .3	0.5	50
10NS	5.0	± .3	0.5	50
11NS	5.5	± .3	0.5	50
12NS	6.0	± .3	0.5	50
13NS	6.5	± .3	0.5	50
14NS	7.0	± .4	0.5	50
15NS	7.5	± .4	0.5	50
16NS	8.0	± .4	0.5	50
17NS	8.5	± .4	0.5	50
18NS	9.0	± .5	0.5	50
19NS	9.5	± .5	0.5	50
20NS	10.	± .5	0.5	50

* Duty cycle 5 x 10^{-4}, maximum pulse duration 5 μsec.

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TABLE I. Electrical characteristics - Continued.

Dash number	Attenuation (dB)		Maximum power input	
	Nominal	Deviation (DC - 18 GHz)	AV(W) at 25°C (continuously)	Peak (*) (W)
21N S	10.5	± .5	0.5	50
22N S	11	± .6	0.5	50
23N S	11.5	± .6	0.5	50
24N S	12	± .6	0.5	50
25N S	12.5	± .6	0.5	50
26N S	13	± .7	0.5	50
27N S	13.5	± .7	0.5	50
28N S	14	± .7	0.5	50
29N S	14.5	± .7	0.5	50
30N S	15	± .8	0.5	50
31N S	15.5	± .8	0.5	50
32N S	16	± .8	0.5	50
33N S	16.5	± .8	0.5	50
34N S	17	± .9	0.5	50
35N S	17.5	± .9	0.5	50
36N S	18	± .9	0.5	50
37N S	18.5	± .9	0.5	50
38N S	19	±.1.0	0.5	50
39N S	19.5	±.1.0	0.5	50
40N S	20.	±.1.0	0.5	50

* Duty cycle 5 x 10^{-4}, maximum pulse duration 5 µsec.

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Part or Identifying Number (PIN): M3933/27- (and dash number from table I).

Referenced documents. In addition to MIL-DTL-3933, this document references the following:

AMS-5640
ASTM-B196/B196M
ASTM-B488
ASTM-D1710
AMS-QQ-N-290
AMS-QQ-P-35
MIL-STD-348

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5985-1320)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/.

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